

frontal plane elevation — random spline, per-cell df by BIC, D diagonal

Sternoclavicular — Protraction(+)/retraction(-)

Sternoclavicular — Depression(+)/elevation(-)

Sternoclavicular — Backwards(+)/forward(-) rot



Acromioclavicular — Protraction(+)/retraction(-)

Acromioclavicular — Medial(+)/lateral(-) rot

Acromioclavicular — Posterior(+)/anterior(-) tilt



Scapulothoracic — Protraction(+)/retraction(-)

Scapulothoracic — Medial(+)/lateral(-) rot

Scapulothoracic — Posterior(+)/anterior(-) tilt



Glenohumeral — Plane of elevation

Glenohumeral — Elevation(-)/Depression(+)

Glenohumeral — Int(+)/ext(-) rotation



orange = ex vivo, green = in vivo · grey = one pooled curve (cell not compared) · line = population curve, band = 95% CI. No knots are drawn.

stars = LIKELIHOOD-RATIO TEST of H0 'no ex-vivo departure' against H1, Bonferroni-adjusted across every compared cell on the plate: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, ns = not significant.

ex/in · studies ex/in | K , this cell's own spline df chosen by BIC | mean and max $|\Delta|$ in degrees. Curves span the full abscissa — every x any shoulder reaches is fitted, so the thinnest stretches rest on a single shoulder and the 95%